



## **Attachment 2: Questions to Accompany the Explosives Trace Detection Request for Information**

### **General Systems Description:**

1. Describe the system's operating principle(s).
  - a. Provide summary of enabling technology or technologies employed.
  - b. Provide block diagrams/schematic illustrating major system components and functionality.
  - c. Does the system require the use of consumable items? If so, please describe what those items are and frequency of replacement?

### **Detection and False Alarm Rates:**

2. Provide a list of explosives and chemicals the system detects and their limits of detection.
3. Does the system have the capability to add additional explosives and chemicals to its library of detectable threats using a simple means such as loading a new detection algorithm?
4. Has the system undergone an independent test and evaluation of the system detection? If yes, please provide any test reports available.
5. What is the False Alarm Rate?

### **Processing Time/Throughput:**

6. What is the Machine Processing and Analysis Time? Machine Processing and Analysis Time is defined as beginning when a trace sample is introduced to the system and ends when the system displays the scan results.
7. What is the average alarm clear-down time? Clear-down time is defined as the amount of time it takes for the machine to ready itself for a new sample after an alarm.

### **Start-up:**

8. How long is the machine's cold start-up procedure from power initiation to display of the login screen? Cold start-up occurs when the system has been turned off for a period of thirty (30) minutes or more and then is turned on.
9. How long is the system warm start-up procedure from power initiation to display of the login screen. Warm start-up occurs when the system has been turned off for a period of less than thirty (30) minutes and then is turned on.

### **Audible and Visual Indicators:**

10. Does the system provide any audio or visual indicators?
  - a. Upon detection/non-detection of a threat.
  - b. When the system is ready to accept a sample.

### **System Error and Faults:**

11. Does the system allow sample analysis when an error or fault is remedied?

### **Calibration/Verification Check:**



12. Does the system perform a calibration/verification or prompt the user to perform a calibration/verification to ensure the system is calibrated properly?
13. If the system prompts the user to perform a calibration/verification, at what intervals does the system perform a calibration/verification or prompt the user to perform a verification?
14. How is the calibration/verification of the system performed?

**Field Data Reporting and Recording:**

15. Does the system include a Field Data Reporting System (FDRS) that can collect, analyze, store and output data?
16. Does the system FDRS accurately record and report sample event data? Sample event data is defined as all relevant information that can be associated with a sample analysis. At a minimum, the relevant information includes a timestamp and an encoded identifier for the detected explosives and/or chemicals. Additionally, it may include such information as explosives and /or chemicals signal strength or plasmagrams.

**Error and Fault Messages:**

17. When a system error or fault occurs, does the system provide an audible indicator and display a notification message on the display monitor?
18. Does the system track error and fault messages in the FDRS?

**Data Storage and Transfer:**

19. How many sampling events can the system store? Sample event data is defined as all relevant information that can be associated with a sample analysis. This can include such information as explosives and/or chemicals signal strength or plasmagrams.

**Physical:**

20. What are the dimensions of the system?
21. What is the weight of the system?

**Printer:**

22. Does the system provide a printing means to print collected data and test data to support routine operations?

**Reliability:**

23. What is the Mean Time Between Critical Failures (MTBCF)?  
For determining the reliability of Transportation Security Equipment (TSE) in the operational environment, MTBCF considers the System Operating Time/Hours and the Number of Critical Failures. System Operating Time is the period a system is available to perform its required mission. Number of Critical Failures is the number of failures which prevents the TSE from performing its intended function and cannot be cleared by operator intervention alone; the fault requires a call to the maintenance service provider to restore the system to operation. Events that a Transportation Security Officer (TSO) can correct via system resets, simple removal of obstructions, etc., are not categorized as critical failures. MTBCF is calculated as follows:



$$\text{MTBCF} = \frac{\text{System Operating Time}}{\text{Number of Critical Failures}}$$

### Maintainability:

24. What is the Mean Down Time (MDT)?

For determining the Maintainability of TSE for calculating Availability in the operational environment, the TSA considers the total operational time that a TSE is not in a condition to perform its required mission. This includes the time from the initial notification of the critical failure until the time the TSE is ready for acceptance and signoff by the TSA.

Mean Down Time (MDT) is calculated by dividing the total operating time that a system is not in a condition to perform its required mission by the number of failures over a given period.

$$\text{MDT} = \frac{\text{Total Operating Time the System is Down}}{\text{Number of Failures Over a Given Period}}$$

### Availability:

25. What is the system Availability?

Availability (Ao) is the percentage of time, over a period during operational hours, that TSE is available to perform its required mission. It reflects all potential causes of downtime, such as administrative downtime, logistics downtime, etc., and total TSE operational uptime. Ao is calculated as follows:

$$\text{Ao} = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}} \times 100$$

### Safety:

26. Does the system comply with *OSHA Regulations 1910.303(a) and 1910.303(b), Electrical – General Requirements*, and *UL 61010, Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use – Part 1: General Requirements*?

27. Does the system have or use any hazardous materials such as chemical(s), radioactive sources, lasers, etc.? If yes, what type of protective equipment will be required for the operation and maintenance of the machine?

28. Does the system expose users or maintenance personnel to extreme temperatures?

29. Does the system expose users to ergonomic risk factors from awkward postures, stresses, and forces during system operation, calibration, and maintenance?